

Optimizing neural network architecture using metaheuristic algorithms

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Overview

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Introduction - Problem Description

- **Given** an image-based dataset with n data points $d^{(1)}, d^{(2)}, \dots, d^{(n)} \in \mathbb{R}^{H \times W \times C}$, where H, W, C represent height, width and channels respectively, and $n \in \mathbb{N}$ represents the number of samples.
- **Find** $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathcal{D}^n$ which minimizes

$$f(\alpha, \mathcal{D}) := \lambda_1 \cdot \text{NetworkSize}(\alpha) - \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D})$$

Where:

- α represents the network architecture parameters
- $\mathcal{D}$ is the dataset
- λ_1, λ_2 are weighting factors balancing size vs. accuracy
- $\text{NetworkSize}(\alpha)$ counts the total number of parameters
- $\text{Accuracy}(\alpha, \mathcal{D})$ is the classification accuracy on dataset $\mathcal{D}$

- **Motivation** (Optimization of NNA):

- Non-linear cost function $f() =$
- Discrete / Continuous solution space $D_s = \{\}$
- TBD
- Solution space is represented by soggy cat
- **Naive Approach:** Brute-Force Algorithm with run time $\mathcal{O}(2^x)$ (x still has to be determined)

Introduction - Project Objectives

- Investigate how well the **Genetic Algorithm** (normal and enhanced) and **Particle Swarm Optimization** perform when applied to the problem.
- TBD
- TBD
- Assess the performance of these algorithms by comparing them to the **Brute-Force** and **Monte-Carlo** approach.

Algorithms - Brute Force and Monte-Carlo

Algorithms - Simulated Annealing

Algorithms - (Enhanced) Genetic Algorithm

Algorithms - Particle Swarm Optimization

Parametric Optimization

Parametric Optimization

Parametric Optimization - Genetic Algorithm

Results

Conclusion and Discussion

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